

# Revathy Menon

PhD Candidate · Computational Structural Biology · NCBS-TIFR, Bangalore

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Computational biology PhD researcher (thesis submitted) specialising in protein–protein interactions, ML-augmented docking, and structural bioinformatics. 6 peer-reviewed publications (3 co-first author). CSIR-NET AIR 65 (top 0.1%).

## EDUCATION

**PhD, Biological Sciences — National Centre for Biological Sciences (NCBS-TIFR)** 2021 – Present

*Supervisor: Prof. Ramanathan Sowdhamini | CGPA: 7.9/10 | Grade A at all thesis committee meetings*

*Passed Comprehensive Examination (Part II) on first attempt, Nov 2023*

**Integrated BS-MS, Biology — IISER Tirupati** 2016 – 2021

*CGPA: 7.4/10 | MS Thesis: Bifurcated inter-protein interactions in protein complexes (IISc / IISER Tirupati) — co-first-author publication*

## FELLOWSHIPS & AWARDS

**DAE-TIFR Junior/Senior Research Fellowship** (via JGEEBILS) — full doctoral funding 2021–Present

**CSIR-NET JRF — All India Rank 65** (top 0.1% nationally) 2019

**DST INSPIRE Scholarship for Higher Education** — top 1% nationally 2016–2021

**Infosys Foundation Travel Grant** — ₹50,000 for EMBL AI & Biology Conference, Heidelberg 2026

**NTSE Scholarship** — Ministry of Education (NCERT) 2013–2016

## RESEARCH EXPERIENCE

**PhD Researcher — NCBS-TIFR, Bangalore** 2021 – Present

- **DockScore 2.0:** Re-implemented and extended a protein–protein docking scorer in Python. The scorer is a two-stage cascade. The first XGBoost stage predicts CAPRI quality and filters out predicted incorrect decoys. The second GNN stage ranks the surviving decoys. Benchmarked on CAPRI Scoreset v2022 across 257 complexes and 100K+ decoys. Poster at *EMBL AI and Biology Conference, Heidelberg (2026)*.
- **PASS2 v8 Database:** Led update of structure-based sequence alignment database covering 26,690 protein domains across 2,058 superfamilies (SCOPe 2.08). 90.4% pipeline success; recovered 79 problematic superfamilies via k-means clustering. *Published — Database, 2025*.
- **Actin–Tropomyosin Interface:** Modelled actin–tropomyosin co-polymer interfaces across human (hTPM1, hTPM4) and *S. pombe* (CDC8) isoforms using AlphaFold2-Multimer + LightDock docking. Identified conserved electrostatic anchors and a "gestalt-binding" energy hierarchy. Collaboration with Univ. of Warwick. *Preprint — bioRxiv, 2026*.
- **TRAM Dimer Mutations:** 200 ns MD simulations (×3 replicates, GROMACS) + allosteric path analysis revealing structural basis for abrogated innate immune signalling. *Published — Proteins, 2024 (top 10% most-viewed)*.
- **RNA-seq Pipeline:** End-to-end bulk RNA-seq analysis (FastQC → HISAT2 → StringTie → DESeq2) for transcriptomic profiling in the L-DOPA pathway.

**Pre-Doctoral Research Internships** 2017 – 2021

- **IISER Tirupati / IISc (2020–21):** MS Thesis — Bifurcated inter-protein interactions. Supervisors: Prof. N. Srinivasan & Dr. S. Chavali.
- **NCBS, Bangalore (2018):** Trait-demography relationships in S. Indian tropical forests; Western Ghats fieldwork. Supervisor: Dr. Mahesh Sankaran.
- **IMSc, Chennai (2017):** Modelling dengue vectors *Aedes aegypti* and *A. albopictus*. Supervisor: Dr. Gautam Menon.

## PUBLICATIONS (\* CO-FIRST AUTHORSHIP)

1. Joshi S.\*, Pathak A.\*, Regati D.R.\*, **Menon R.**, et al. GRAYU: graph-based database integrating Ayurvedic formulations, medicinal plants, phytochemicals and diseases. *Frontiers in Pharmacology*, 16:1727224. doi: 10.3389/fphar.2025.1727224 (2026)
2. **Menon R.**\*, Bheemireddy S.\*, Chavali S., Sowdhamini R., Srinivasan N. Computational analyses of bifurcated inter-protein interactions reveal their pivotal role in conferring stability. *Proteins: Structure, Function, and Bioinformatics*. doi: 10.1002/prot.70090 (2025)
3. **Menon R.**\*, Nayak S.\*, Rajesh R., Sowdhamini R. PASS2: update of database of structure-based sequence alignments. *Database*. doi: 10.1093/database/baaf072 (2025)
4. Verma S., Sharma A., Pathak A., **Menon R.**, Sowdhamini R. Evolution of Protein Families. *Encyclopedia of Bioinformatics and Computational Biology (2nd ed.)*, 3:99–110 (2025)
5. Pathak A., Tiwari V., **Menon R.**, Sowdhamini R. Prediction of Protein–Protein Interactions. *Encyclopedia of Bioinformatics and Computational Biology (2nd ed.)*, 4:380–396 (2025)
6. **Menon R.**\*, Verma S.\*, Sowdhamini R. Structural insights into deleterious mutations at the TRAM dimeric interface. *Proteins: Structure, Function, and Bioinformatics*. doi: 10.1002/prot.26707 (2024)

## TEACHING & MENTORSHIP

### Graduate Teaching Assistant — NCBS-TIFR

2022 – 2024

- *Statistical Inference in Biology* (2022 & 2023): Tutorials on MCMC, Bayesian inference, SVMs, neural networks; supervised 10+ student ML projects (30–40 students/iteration).
- *AI for Good and Evil — Generative AI* (2024): Transformers, LLMs, diffusion models, prompt engineering, AI ethics for mixed STEM/non-STEM cohort.
- *Guest Lecturer — AI in Structural Biology* (2025): AlphaFold 2/3, history of structure prediction, live PyMol + ColabFold demos.
- Mentored 8 undergraduate/project students across 4 years on structural analysis, docking, and phylogenetics.

## TECHNICAL SKILLS

**Programming:** Python, R, SQL/MariaDB, PHP, HTML/CSS, LaTeX, Django

**ML / Statistics:** XGBoost, Random Forest, SVM, Neural Networks, GNNs, Knowledge Distillation, MCMC, LMER, DESeq2

**Libraries:** scikit-learn, NumPy, Pandas, Matplotlib, BioPython

**Structural Biology:** AlphaFold 2/3, GROMACS, LightDock, HADDOCK, PyMol, FoldX, PPCheck, ColabFold, HMMER, BLAST

**Genomics:** HISAT2, StringTie, FastQC, DESeq2, edgeR

**Other:** Budget management (₹30–40L/yr), vendor negotiation, audit compliance, stakeholder reporting

## BEYOND RESEARCH

- **Science Communication:** Co-founder & Managing Editor, *Inventa* (inter-institute sci-comm initiative, 2021); Chief Editor, *Dhwani* (IISER Tirupati magazine, 2019–20); NCBS Research Explained web editor; scientific illustrations published in a peer-reviewed mathematics journal (2018) and exhibited at NCBS Open Day (2024).
- **Conferences:** Poster — EMBL AI and Biology Conference, Heidelberg (March 2026); NCBS Annual Talks (2025); Annual Institute Research Seminars (2023, 2024, 2025).
- **Workshops:** Summer School on AI/ML, IIT Tirupati (2019); Physics of Life — NCBS-Simons Monsoon School (2018); National Science Camp (Vijyoshi), IISc (2016).